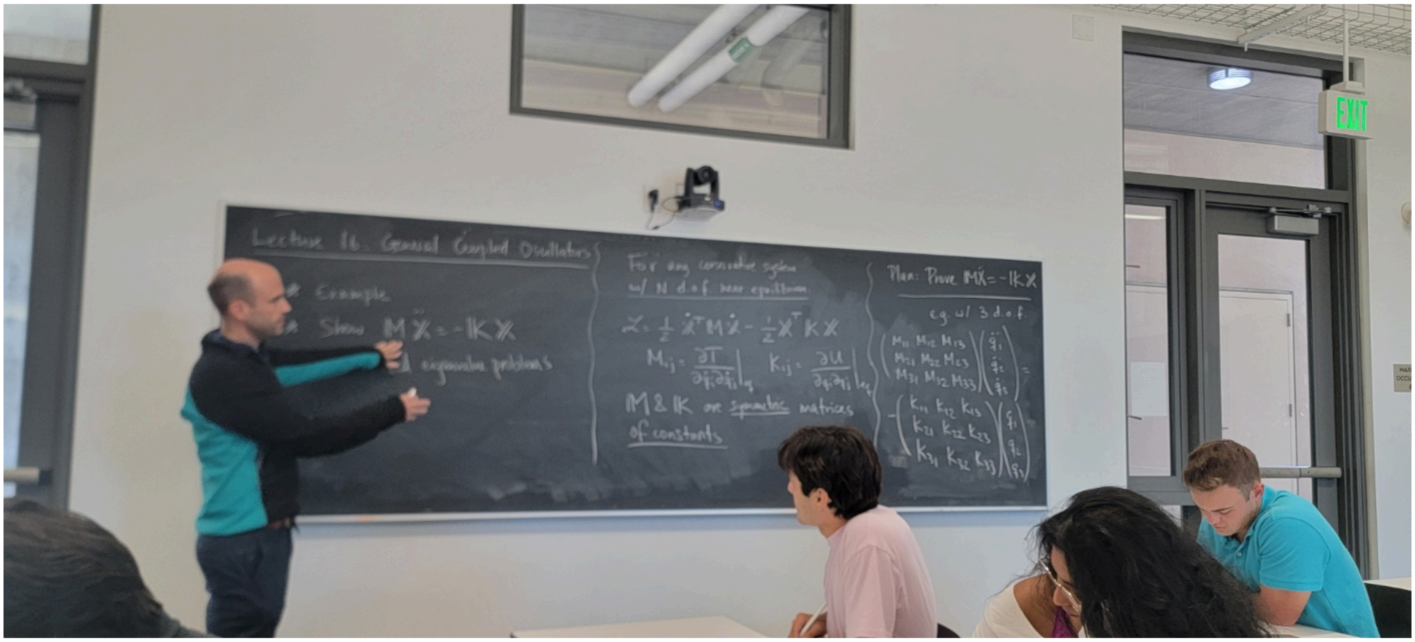
15 Normal Modes pt 3 Linear Algebra

Review

For any conservative system with N degrees of freedom, we can write the Lagrangian as

$$\mathscr{L} = \frac{1}{2}\dot{\mathbb{X}}^T \mathbb{M} \dot{\mathbb{X}} - \frac{1}{2}\mathbb{X}^T \mathbb{K} \mathbb{X}$$

Where



where

$$\mathbb{M}_{ij} = \frac{\partial T}{\partial \dot{q}_1 \partial \dot{q}_j}\bigg|_{\text{equilibrium}}$$
$$\mathbb{K}_{ij} = \frac{\partial U}{\partial q_1 \partial q_j}\bigg|_{\text{equilibrium}}$$

These matrices should be symmetric, since the derivatives commute. Note that we're expanding around the equilibrium when evaluating each of these partials.

For example, the Lagrangian

$$\mathscr{L} = \frac{M}{2}(\dot{x}_1^2 + \dot{x}_3^2) + \frac{m}{2}\dot{x}_2^2 - \frac{k}{2}(x_3 - x_2)^2 - \frac{k}{2}(x_2 - x_1)^2$$

Here,

$$\mathbb{M} = \begin{bmatrix} M & 0 & 0 \\ 0 & m & 0 \\ 0 & 0 & M \end{bmatrix}$$

and

$$\mathbb{K} = \begin{bmatrix} k & -k & 0 \\ -k & 2k & -k \\ 0 & -k & k \end{bmatrix}$$

Note that this is not $-2k$ in two of the entries - when we foil the (stuff)^2 we go really fast and combine terms, but here we're leaving them uncombined because its helpful to be symmetric. Upper triangular is technically right, but who cares.

Our $\mathbb{X} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

Lecture

We will prove that this matrix Lagrangian yields a differential equation which looks like

$$\mathbb{M}\ddot{\mathbb{X}} = \mathbb{K}\mathbb{X}$$

In index notation,

$$\mathscr{L} = \frac{1}{2}\sum_{i,j}\dot{q}_i M_{ij}\dot{q}_j - \frac{1}{2}\sum_{i,j} q_{iij} q_j$$

We can write down the Euler Lagrange equation for the n^{th} coordinate. Note that $M_{ij} \text{ and } {}_{ij}$ are both scalers, not matrices - they are the values in the ij^{th} entry of their respective $\mathbb{M}, \mathbb{K}$.

$$\frac{d}{dt}\left(\frac{\partial \mathscr{L}}{\partial \dot{q}_n}\right) = \frac{\partial \mathscr{L}}{\partial q_n}$$

The right side:

$$\frac{\partial \mathscr{L}}{\partial q_n} = -\frac{1}{2}\sum_{i,j} {}_{ij}\frac{\partial}{\partial q_n}(q_i q_j)$$

We can break this out into two sums

$$-\frac{1}{2}\sum_{ij} {}_{ij}\left(\frac{\partial q_i}{\partial q_n} q_j + q_i \frac{\partial q_j}{\partial q_n}\right)$$

Note that this term inside acts like the Kronecker delta - its going to be a 1 if $q_i = q_n$, and 0 otherwise. This isolates out terms where $i = n$ or $j = n$. We have

$$-\frac{1}{2}\sum_{ij} {}_{ij}\left(\frac{\partial q_i}{\partial q_n} q_j\right) - \frac{1}{2}\sum_{ij}\left(q_i \frac{\partial q_j}{\partial q_n}\right)$$

Which becomes

$$-\frac{1}{2}\sum_{ij} {}_{ij}\delta_{in} q_j - \frac{1}{2}\sum_{ij} {}_{ij}\delta_{jn} q_i$$

This gives us only the terms where $i = n$ or $j = n$.

$$-\frac{1}{2}\sum_{j} {}_{nj} q_j - \frac{1}{2}\sum_{i} k_{in} q_i$$

(side note - this is symmetric because we have a symmetric matrix! One we're summing along rows, another along columns).

Because ${}_{nj} = {}_{jn}$, and because our labeling is arbitrary, we can just write this out

$$\begin{aligned} -\frac{1}{2}\sum_{j} {}_{nj} q_j - \frac{1}{2}\sum_{j} k_{jn} q_j \\ = -\frac{1}{2}\sum_{j} {}_{nj} q_j - \frac{1}{2}\sum_{j} k_{nj} q_j \\ = -\sum_{j} {}_{nj} q_j \end{aligned}$$

Remember, we can write out

$$\mathbb{K} = \begin{bmatrix} {}_{11} & {}_{1,2} & {}_{1,3} \\ & & \\ {}_{3,1} & & {}_{3,3} \end{bmatrix}$$

and

$$\mathbb{X} = \begin{pmatrix} q_1 \\ q_2 \\ q_3 \end{pmatrix}$$

This means that the nth entry in our Lagrange equation will have a right side that looks like

$$(\mathbb{K}\mathbb{X})_n$$

(the nth entry down of this)

The left side of the Euler Lagrange equation is

$$\frac{\partial \mathscr{L}}{\partial \dot{q}_n} = \sum_j M_{n,j}\dot{q}_j = (\mathbb{M}\ddot{\mathbb{X}})_n$$

We now have a **generalized eigenvalue problem.**

The strategy is similar to before - we'll move to complex variables, and guess that $\mathbb{Z}(t)$ looks like $\mathbb{Z}_0 e^{i\omega t}$.

Lets see what parts are the same, and which are unique:

$$\begin{aligned} \mathbb{M}\mathbb{Z}_0(-\omega^2)e^{i\omega t} &= -\mathbb{K}\mathbb{Z}_0 e^{i\omega t} \\ (\mathbb{K} - \omega^2\mathbb{M})\mathbb{Z}_0 &= 0 \\ \det(\mathbb{K} - \omega^2\mathbb{M}) &= 0 \end{aligned}$$

We can solve for the eigenvalues, and then get the corresponding eigen vectors that are the normal modes for our system, and then write a general solution as a sum over all the eigenmodes.

We can show that the eigenvalues ω^2 must be real and positive (no imaginary ω), and that the eigenvectors are orthogonal, although with a slightly different definition of the inner product.

$$\mathbb{Z}_0^{(i)^T}\mathbb{M}\mathbb{Z}_0^{(j)} = 0 \text{ (if i } \neq \text{)}$$

We can always write our general solution

$$\mathbb{Z}(t) = \sum_{i=1}^{N} C_i \mathbb{Z}_0^i e^{i\omega t}$$

Where

- N is number of of degrees of freedom, which is the same as the number of normal modes,
- C_i is the complex amplitude of mode i, amplitude and phase.
- $\mathbb{Z}_0$ is the eigenvector that tells you how much of mode i is in each q_i (i.e. $\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$)

  We are adding the contribution from all normal modes to all variables to find how they evolve over time.